ASAD HUSSAIN

asadh1521@gmail.com · +92 320 4141092 · Portfolio Website . Lahore, Pakistan · Linkedin|Asad Hussain

PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at **NUST** with a multidisciplinary focus on **Artificial Intelligence, Machine Learning, Embedded Systems, and RF Engineering**. Experienced in applied ML and AI engineering through **FlyRank AI, CodeAlpha, and atomCamp**, with hands-on exposure to **LLMs, RAG, deep learning, and backend development**. Complementing software expertise with practical experience in **RF hardware, embedded firmware, IoT, and signal-processing systems**, including **DroneGuard**, a two-board RF detection . Adept at bridging **AI-driven software with real-world hardware and sensor data**, from model development and data processing to circuit design and embedded implementation

TECHNICAL SKILLS

Programming: Python · C++/Embedded C · JavaScript · SQL · MATLAB/Simulink · Flask
ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM Fine-tuning · FastAPI · Docker
Hardware & RF: SDR · STM32 · Arduino · Raspberry Pi · PCB Design (Proteus) · Antenna Design
Tools: Git · REST APIs · SQL Server · Google Cloud · Weights & Biases · AutoCAD · Altium Designer

WORK EXPERIENCE

Technical Intern | National Electric Power Regulatory Authority (NEPRA), *Islamabad* — **Jul – Aug 2026**

- Studied NEPRA's end-to-end regulatory flow (license → tariff → sale) across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms.
- Analyzed NGC's transmission network data, 233 grid-station transformers (116 above 80% rated capacity), and NEPRA's State of the Industry Report, covering 111,466 GWh of national consumption.

Backend AI Engineering Intern | FlyRank AI, *Remote, Chicago, USA* — **Jun – Aug 2026**

- Completed a backend/AI engineering practicum: built an authentication-protected CRUD API with a connected database, containerized the stack with Docker, and automated a data-scraping/workflow pipeline.
- Applied prompt-engineering and LLM-tooling fundamentals to search-intelligence tasks (baseline scoring and top-10 review pipeline) across 21 practical assignments and 19 workshops on AI tooling and backend development.

Machine Learning Intern | CodeAlpha, *Remote* — **May – Jun 2026**

- Built a credit-scoring classification model (Logistic Regression, Decision Trees, Random Forest) with feature engineering from financial history data, evaluated using Precision, Recall, F1-score, and ROC-AUC.
- Built a CNN-based handwritten character recognition model (MNIST/EMNIST) and a disease-prediction classifier on structured medical data (symptoms, age, lab results) using SVM, Logistic Regression, and Random Forest.

Engineering Intern, Full-Stack Systems | Research & Indigenous Development Centre (RDC), HIT — **Jun – Aug 2025**

- Integrated core hardware (LilyGO T-SIM7670E, T3-S3 GPS, RDM6300 RFID) with C/C++ state machines for GPS capture, RFID authentication, and MQTT telemetry over cellular GPRS.
- Built an asynchronous Python Flask backend with multithreading to process analytics workloads without blocking incoming MQTT streams.

PROJECTS

DroneGuard — Anti-Drone Surveillance System — **2025 – 2026**

SDR · GNU Radio · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: a four-band RF detection front-end (433 MHz–2.4 GHz) and a jamming board built around an ADF4351 PLL synthesiser.
- Built a real-time signal-classification pipeline (RTL-SDR + GNU Radio) identifying FHSS, OFDM, and FSK drone protocols from live IQ samples, plus a MATLAB front-end for live spectrum and telemetry.

Academic Projects

- Hands-on projects across core EE disciplines: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus) · Current Transformer

TRAINING

- AtomCamp AI Bootcamp — Deep Learning, NLP/LLMs, n8n, Generative AI & RAG (LangChain agent development), and MLOps/deployment (Docker, FastAPI, CI/CD, Google Cloud).

EDUCATION

B.E. Electrical Engineering | National University of Sciences and Technology (NUST) — **2023 – 2027**

- Coursework: Analog Electronics, DSP, Control Systems, Communication Systems, Digital Logic Design, Embedded Systems, Signals & Systems, Instrumentation, OOP, Microprocessor Systems

CERTIFICATIONS, ACTIVITIES & ACHIEVEMENTS

- CS50x, **Harvard University** (2026) · Develop AI Apps and Agents on Azure (AI-103T00-A), **Microsoft/edX** (2025) · Data Analysis with Python, **IBM (2025)** ,PCB Basic Design, Altium Education (2026) · Effective Leadership, **Deloitte WorldClass** scored 90% (2026) · Digital Product Management, Alison (2024)
- Technical Lead, NUST Robotics Society · Highest Achiever, SSC · Voltfest Participant · NMUN 2024 Delegate